

Carbon Capture process utilizing a 30 wt% MEA solvent as modeled in Aspen Plus

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Abstract

Post-combustion CO₂ capture using 30 wt% monoethanolamine (MEA) solvent was simulated using a steady-state rate-based model in ASPEN Plus, validated with pilot data and common literature findings. The simulation employed ASPEN's ELECNRTL property method and rate-based RadFrac columns with FLEXIPAC 250Y structured packing to model the absorber-stripper loop. The results achieved a 90.1% CO₂ capture rate at a reboiler duty of 4.147 GJ/tonne CO₂. Lean and rich solvent loadings were 0.302 and 0.535 mol CO₂/mol MEA respectively, with a liquid-to-gas ratio (L/G) of 2.7. Sensitivity analyses quantified the tradeoff between reboiler duty and lean loading, identified the minimum L/G for 90% capture, and characterized the effect of stripper operating pressure on duty. An optimization block minimized reboiler duty to 178.1 MW while maintaining 91.4% capture. Screening-level techno-economic analysis using ASPEN Economic Analyzer showed a levelized cost of capture of \$56.93/tonne CO₂, consistent with NETL benchmark estimates for first-generation MEA post-combustion capture systems.

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Carbon Dioxide (CO₂) released from the burning of fossil fuels is one of the leading greenhouse gases contributing to global warming. Because of these various strategies to reduce carbon emissions are being adapted across multiple industries. One of the most promising strategies that has emerged is carbon capture. Carbon capture seeks to trap carbon emissions released from the burning of gas or coal and create a stream of pure carbon that can be contained and returned to the earth's mantle or used in other industries (MIT, 2025). This is a viable strategy as this process can be retrofitted to existing power plants.

There are a variety of modern carbon capture techniques utilizing different solvents that follow a generally similar chemical process to achieve a clean gas stream. The solvent that this paper focuses on is monoethanolamine (MEA) due to its consistent use and the breadth of validation data available. The paper 'Amine Scrubbing for CO₂ Capture' describes the process and equipment necessary for carbon capture using MEA. An aqueous solution of MEA is used to absorb carbon dioxide from a plant's flue gas (Rochelle, 2009). The amine is then recycled through a stripper that separates the MEA and carbon with water vapor (Rochelle, 2009). This results in streams of nearly pure water, MEA, and carbon dioxide. This MEA can then be reused to absorb more carbon.

The dominant cost of this process results from the heat duty of the reboiler within the stripping process. The energy from this process represents about half of the work for the entire process with the other half coming from compressing CO₂ for transport (Rochelle, 2009). Therefore, this presents the main challenge in improving modern carbon capture processes. Reboiler duty needs to be minimized to manage costs while still maintaining at least 90% capture which is the U.S. Department of Energy (DOE) benchmark.

This paper will investigate post-combustion-carbon-capture using 30% MEA as a solvent. This will be done using Aspen Plus software to create a steady state rate-based simulation of the standard absorber-stripper loop. The development of this simulation will primarily reference internal Aspen Plus documentation specifically the example model 'Rate-Based Model of the CO₂ Capture Process by MEA using Aspen Plus' and the article 'Optimization of post combustion carbon capture process-solvent selection.' To ensure model credibility data from the simulation will be compared to data from various established literature sources. Data from the simulation will be further analyzed through sensitivity, optimization, and screening level techno-economic analysis using Aspen Plus internal software.

Theory

Post-combustion CO₂ capture with MEA proceeds through a combination of kinetic and equilibrium reactions in the liquid phase. The dominant reaction in the absorber phase of this process is

the kinetic bimolecular reaction between MEA and CO₂ as modeled in Table 2, Reaction 3. This reaction controls the rate of the overall chemical process in the absorber and requires a finite amount of contact time between the gas and liquid phases. Due to this the height and packing of the absorber column directly affects the amount of carbon the MEA is able to capture. Once this reaction occurs, a series of acid-base equilibrium reactions as outlined in Table 1 occur. These reactions control the speciation of CO₂ between carbamate, bicarbonate, and carbonate forms. This process is controlled by the loading and temperature of the simulation. When CO₂ loading is low, as is the case with the lean solvent that enters at the top stage of the absorber, carbamate is the dominant CO₂ containing species. When loading is high free MEA is limited and CO₂ reacts with water as shown in Table 1, Reaction 2, rather than amines.

In the stripper column all equilibrium reactions shift into the reverse direction due to elevated temperatures from the reboiler. At this temperature the carbamate from the absorber breaks down releasing CO₂ allowing for the regeneration of free MEA that is recycled back into the absorber. The temperature at which this regeneration occurs efficiently is controlled by stripper operating pressure. All reactions discussed in this section are summarized in the following two tables which were adapted from Aspen Techs ELENRTL MEA Carbon Capture file (AspenTech, 2014).

Table 1.
Equilibrium Reactions

Reaction	Name	Stoichiometry
1	Water Dissociation	$2 \text{H}_2\text{O} \rightleftharpoons \text{H}_3\text{O}^+ + \text{OH}^-$
2	CO ₂ Hydration	$\text{CO}_2 + 2 \text{H}_2\text{O} \rightleftharpoons \text{HCO}_3^- + \text{H}_3\text{O}^+$
3	Bicarbonate Dissociation	$\text{HCO}_3^- + \text{H}_2\text{O} \rightleftharpoons \text{CO}_3^{2-} + \text{H}_3\text{O}^+$
4	MEA Protonation	$\text{MEA} + \text{H}_3\text{O}^+ \rightleftharpoons \text{MEAH}^+ + \text{H}_2\text{O}$
5	Carbamate Reversion	$\text{MEACOO}^- + \text{H}_2\text{O} \rightleftharpoons \text{MEA} + \text{HCO}_3^-$

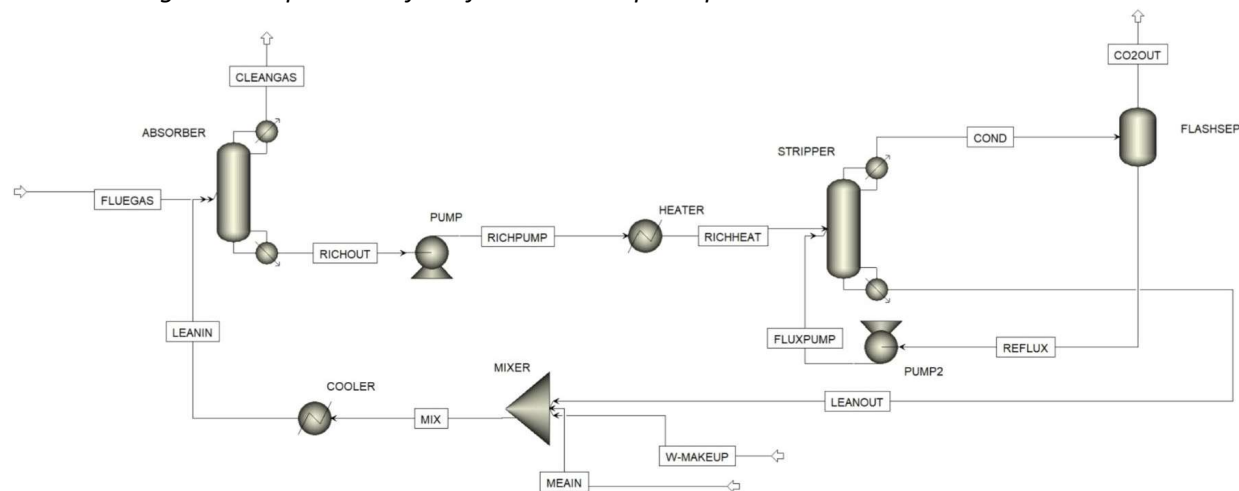
Table 2.
Kinetic Reactions

Reaction	Name	Stoichiometry
1	Base-catalyzed CO ₂ hydration (forward)	$\text{CO}_2 + \text{OH}^- \rightarrow \text{HCO}_3^-$
2	Base-catalyzed CO ₂ hydration (reverse)	$\text{HCO}_3^- \rightarrow \text{CO}_2 + \text{OH}^-$
3	Carbamate formation	$\text{MEA} + \text{CO}_2 + \text{H}_2\text{O} \rightarrow \text{MEACOO}^- + \text{H}_3\text{O}^+$
4	Carbamate decomposition	$\text{MEACOO}^- + \text{H}_3\text{O}^+ \rightarrow \text{MEA} + \text{CO}_2 + \text{H}_2\text{O}$

Simulation

Figure 1.

Flowsheet diagram in Aspen Plus of the full Carbon Capture process



The carbon capture reactions and process outlined above were modeled in ASPEN Plus with RadFrac columns using rate-based calculations. To model the reactions described above the ELENRTL property method was selected. This property method was selected as it allows the simulation to accurately model an ionic amine-CO₂-H₂O system. All reactions, interactions parameters, and activity coefficients were pre-loaded into the properties section of the simulation using the example Aspen Plus file mentioned above (AspenTech, 2014).

The process flow for the simulation, as shown in Figure 1, was slightly modified from traditional carbon capture methods. The FLUEGAS stream enters in the top stage of the absorber block where CLEANGAS and RICHOUT streams are produced. The RICHOUT stream is then sent through a pump and heater block which increase the pressure and temperature of the rich solvent to the STRIPPER's operating specifications. Then the rich solvent enters the top stage of the STRIPPER column which produces a COND and LEANOUT stream. The COND stream is sent to a flash separator to be further processed. The FLASHSEP block separates the CO₂ and water vapor by condensing the water back into liquid stream. This stream is then pumped back up the operating pressure and recycled back into the STRIPPER block. The LEANOUT stream is then sent through a mixer and mixed with a water and MEA makeup stream. This new mixed stream is then cooled back to the absorber columns operating temperature and recycled back into the ABSORBER block. The specific temperature and pressure of each block is described in the table below.

Table 3.

ASPEN block specifications.

Block	Specifications
PUMP	Pressure Increase: 1 bar
HEATER	Temperature: 80 C; Pressure: 3 bar
FLASHSEP	Temperature: 40 C; Pressure: 1.8 atm
PUMP2	Pressure Increase: 1 atm

MIXER	N/A
COOLER	Temperature: 40 C; Pressure: 1.1 bar

Below the final specifications for both RadFrac Columns are shown. These specifications were adapted from two sources “Optimization of... Solvent Selection” (Arachchige et al., 2012) and information from the AspenTech software guide for MEA Carbon Capture modeling (AspenTech, 2014). Specifications were updated and changed to improve convergence and efficiency throughout the process, and the table below reflects the final value selected for each relevant parameter.

Table 4.

Final Column Specifications

Column/Spec	Absorber	Stripper
Stages	15	15
Operating Pressure (bar)	1	2
Re-Boiler	None	Kettle
Condenser	None	None
Packing Type	FLEXIPAC; KOCH; METAL; 250Y	FLEXIPAC; KOCH; METAL; 250Y
Packing Height (meter)	20	18
Packing Diameter (meter)	11	7.5
Reaction Condition Factor	0.9	0.9
Film Discretization Ratio	5	5
Flow Model	Mixed	Mixed
Interfacial area factor	1.5	2
Liquid - Film resistance; Number of discretization points	Discretize film; 5	Discretize film; 5
Vapor - Film resistance	Consider Film	Consider Film
Mass transfer coefficient method	Bravo et al. (1985)	Bravo et al. (1985)
Heat transfer coefficient method	Chilton and Colburn	Chilton and Colburn

Interfacial area method	Bravo et al. (1985)	Bravo et al. (1985)
Holdup Method Correlation	Billet and Schultes (1993)	Bravo et al. (1992)

The simulation was initialized with FLUEGAS representing a exit stream from an industrial sized coal-fired factory. Thus, the stream was set to 500 kg/sec with around 13% CO₂. LEANIN was given an original estimate of 600 kg/sec as a tear stream and was converged around this estimate. The only other streams with initial values given were the MEA and water makeup streams, these initial values were placeholders as two design specifications were used to calculate the actual value needed to maintain mass balance in the simulation. A third design specification was implemented to ensure that the simulation achieved a carbon capture rate of at least 90%. This specification varied the amount of FLUEGAS entering the simulation to find the ideal liquid-to-gas ratio (L/G) to achieve this target. The initial values, specifications, and compositions of these streams is shown in Table 4 and the information of each design specification used is shown in Table 5.

Table 5.

Initial Stream Values

Stream	Flow Rate (kg/sec)	Temperature (K)	Pressure (bar)	Composition (Mole Frac)	
FLUEGAS	500	313	1.1	H2O	0.0818
				CO ₂	0.1358
				N2	0.7286
				O2	0.0354
W-MAKEUP	10	313.15	1.1	H2O	1
MEAIN	10	313.15	1.1	MEA	1
LEANIN (TEAR STREAM)	600	313.15	1.1	MEA	0.175
				H2O	0.7
				CO ₂	0.025

Table 6.*Design Specifications*

Name	Specification	Definition	Vary
CAPTURE	EFF = 0.9 Tolerance = 0.01	$EFF = ("FLUEGAS, CO_2" - "CLEANGAS, CO_2") / "FLUEGAS, CO_2"$	FLUEGAS TOTAL MASS FLOW 100 – 600 kg/sec
MEA	TARGET = 0 Tolerance = 0.001	TARGET = ("MAKEUP, MEA") - ("CLEANGAS, MEA" + "CO2OUT, MEA")	MEAIN TOTAL MASS FLOW 0 – 3 kg/sec
WATER	TARGET = 0 Tolerance = 0.001	TARGET = ("FLUEGAS, H2O" + "MAKEUP, H2O") - ("CLEANGAS, H2O" + "CO2OUT, H2O")	W-MAKEUP TOTAL MASS FLOW 1 – 50 kg/sec

After the simulation had converged at a 90% capture rate, several sensitivity analyses and optimization blocks were run to improve and analyze the simulation. Three sensitivity analyses were conducted; the first compared the capture rate of the simulation to both L/G and the carbon to solvent ratio of the streams entering the absorber column. The second compared the specified reboiler duty of the STRIPPER block to the loading value of the LEANOUT stream as well as the overall capture rate of the simulation. The final sensitivity block compared the specified top stage pressure in the STRIPPER block to the strippers reboiler duty and the overall capture rate of the simulation. After these were conducted an optimization block was ran to minimize reboiler duty while maintaining a 90% capture rate, the specifications for this block are shown in the table below.

Table 7.*Variables for Optimization Block*

Type	Variable	Specifications
Objective	Reboiler Duty (MW)	Minimize
Constraint	Capture Rate	$EFF = ("FLUEGAS, CO_2" - "CLEANGAS, CO_2") / "FLUEGAS, CO_2"$ $EFF \geq 0.9$
Decision	FLUEGAS (kg/sec)	Vary from 200 - 1000 (kg/sec)
Decision	Reboiler Duty (MW)	Vary from 150 - 200 (MW)

Results

The results of this simulation were compared against data from two literature sources to better understand and analyze the results of this simulation. The first source is "Experience with CO₂ capture from coal flue gas in pilot-scale..." which gives pilot data from carbon capture plant erected in Denmark (Knudsen et al., 2009). The second source is "Modeling of CO Capture by Aqueous Monoethanolamine" which similarly modeled a post combustion carbon capture process in ASPEN Plus using RadFrac columns (Freguia & Rochelle, 2003). Key metrics from this simulation including CO₂ capture rate, reboiler duty, and lean/rich loading are compared to the results found in these papers in the table below.

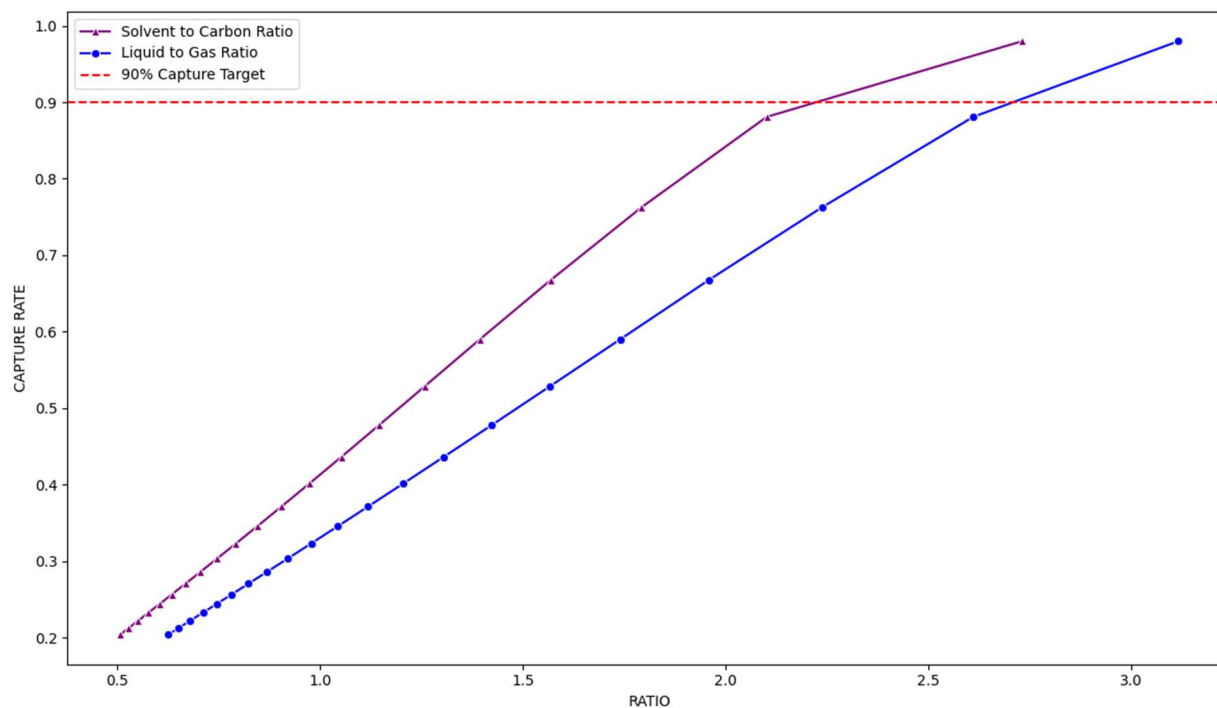
Table 8.
Validation Results

Metric	Source	Validation Target	This Simulation	% error
CO ₂ capture rate	(Knudsen et al., 2009)	90%	90.14%	0.14%
Reboiler Duty (GJ/tonne CO ₂)	(Knudsen et al., 2009)	3.7	4.147	12.08%
Lean loading (mol CO ₂ / mol MEA)	(Freguia & Rochelle, 2003)	0.2 - 0.27	0.3	11%
Rich loading (mol CO ₂ / mol MEA)	(Freguia & Rochelle, 2003)	0.4 - 0.45	0.53	17.77%

The results from this simulation generally trend a little higher than the results from literature and pilot plant data. These deviations can be attributed to differences in column geometry, packing specifications, and various other specific plant details that could not be exactly matched in this simulation. Despite these deviations, the simulation captures the correct directional trends and order-of-magnitude performance, and the CO₂ capture rate of 90.14% meets the DOE benchmark.

Figure 2.

Plot showing how increasing the ratio of MEA solvent to Carbon Dioxide affects the capture rate of the simulation. Target 90% Capture rate is shown in red.

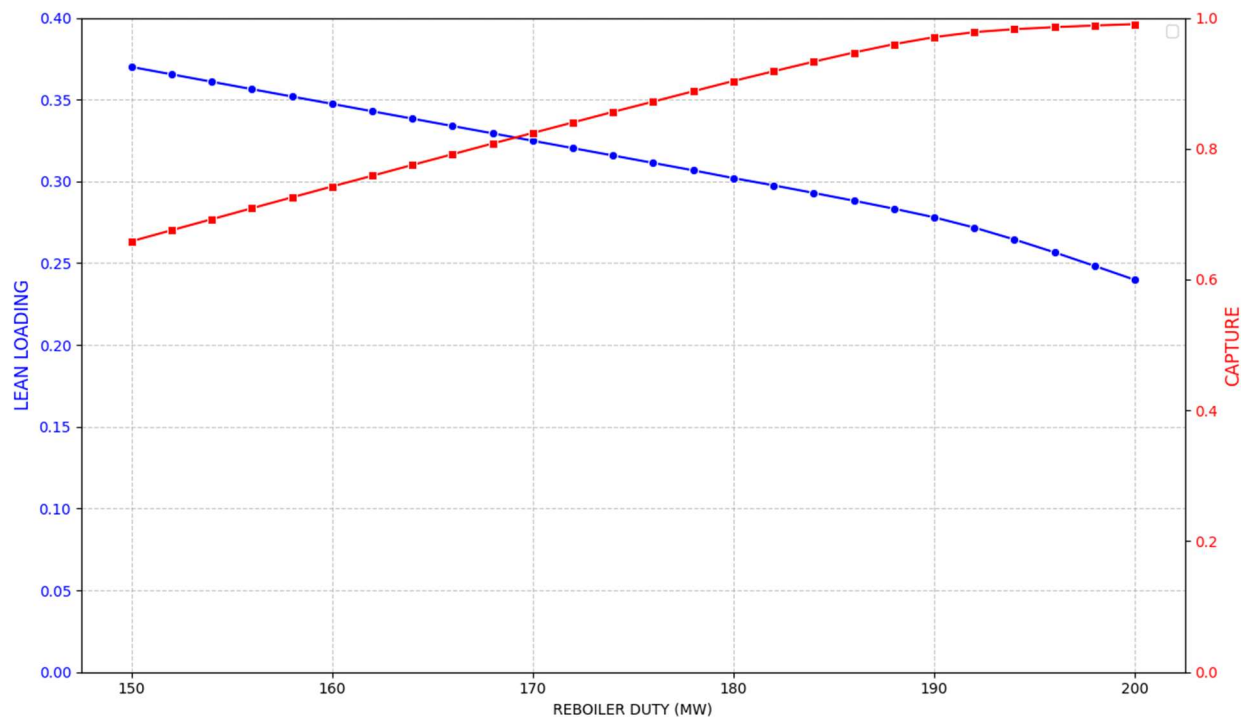


The results of the first sensitivity analysis are shown in the figure above. These results demonstrate that as both the solvent to carbon ratio and L/G increased, the capture rate of the simulation increased as well. The figure shows a sharp positive correlation between the capture rate and

the ratios until the 90% benchmark is reached where the growth slows considerably. The ratios that resulted in a 90% capture rate was around a 2.2 carbon to solvent ratio and a 2.7 L/G.

Figure 3.

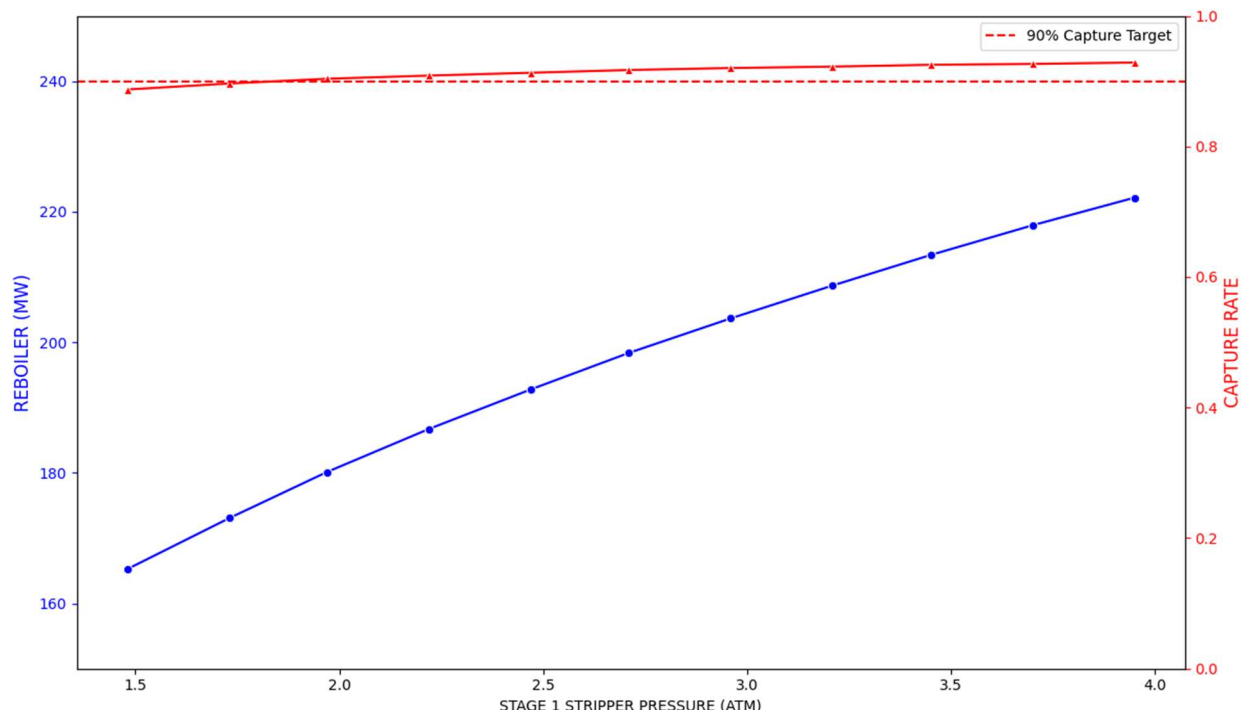
Two axis plot showing the how lean loading and capture rate changes when the reboiler duty of the Stripper is manipulated.



The results of the second sensitivity analysis are shown in figure three. The first trend in the figure is that the value of lean loading decreases as the reboiler duty of the stripper increases. The second trend these results demonstrate is that as the duty increases so does the capture rate. At the optimal condition of a 90% capture rate, the reboiler duty was 180 MW and the lean loading was 0.302.

Figure 4.

Two axis plot showing the how reboiler duty and capture rate changes when the operating pressure of the Stripper is manipulated.



The results of this sensitivity analysis demonstrate how the STRIPPER pressure affects reboiler duty. The figure demonstrates that as the pressure of the stripper column increases the reboiler duty does as well. The figure also demonstrates how the capture rate of the simulation changes with the stripper pressure. The capture rate increases slightly with the pressure change but is not greatly affected. The capture rate also remained above the 90% benchmark for the majority of the analysis. In addition to these analyses an optimization analysis was performed to minimize reboiler duty while maintaining a 90% capture rate. The details of the optimization block are outlined in the table below.

Table 9.

Variables for Optimization Block

Type	Variable	Specifications
Objective	Reboiler Duty (MW)	Minimize
Constraint	Capture Rate	$EFF = \frac{(\text{"FLUEGAS, CO}_2\text{"} - \text{"CLEANGAS, CO}_2\text{"})}{\text{"FLUEGAS, CO}_2\text{'}}$ $EFF \geq 0.9$
Decision	FLUEGAS (kg/sec)	Vary from 200 - 1000 (kg/sec)
Decision	Reboiler Duty (MW)	Vary from 150 - 200 (MW)

The block varied both reboiler duty and the L/G of the simulation. The block used efficiency as a constraint to maintain a 90% capture rate in the simulation. The results of this optimization block were

able to minimize reboiler duty to 178.101 MW while maintaining a capture rate of 0.914. The table below shows the key results of the simulation when it was run at these optimal conditions.

Table 10.

Stream summary when run at optimal conditions: >90% Capture Rate; Reboiler Duty Minimized

PARAMETER	VALUE
Reboiler Duty (MW)	178.101
Capture Rate	0.914
L/G (mol liquid / mol gas)	2.68
Lean Loading (mol CO ₂ / mol MEA)	0.308
Rich Loading (mol CO ₂ / mol MEA)	0.535

There is very little deviation from the results of the simulation when reboiler duty is not minimized. These results demonstrate that the simulation was able to minimize reboiler duty while still maintaining the same level of performance it showed before the minimization. The results of this optimization block were able to decrease reboiler duty by around 2 MW with a 91.14% capture rate.

Techno-Economic Analysis

The techno-economic analysis for this simulation was performed using ASPEN's in house economic evaluation tools. This analysis was intended to be used at a screening level to estimate the possible costs of the carbon capture set up outlined in this simulation. From this data several economic benchmarks were calculated including operating costs, capital costs, and the levelized cost of capture. It must be noted that there was a slight error in the calculations as the reboiler duty was outside the estimation range. Therefore, the actual cost of this plant would most likely be somewhat higher than what is shown here due to this. The economic assumptions made to calculate these benchmarks are outlined in the table below. These assumptions were derived from Aspens Economic tool, literature, and standard economic assumptions in chemical engineering.

Table 11.

Techno-Economic Analysis Assumptions

Parameter	Value	Source
Steam price (\$/GJ)	6.53	ASPEN
MEA price (\$/kg)	1.04	Businessanalytiq, 2020
Plant capacity (tonne CO ₂ /day)	3348.14	ASPEN
Plant lifetime (years)	20	Standard TEA assumption
Discount rate	10%	NETL baseline

Table 12.*Calculated Annual Economic Benchmarks; Assumed 8000 operating hours per year*

Parameter	Value
Operating (\$USD/year)	67981463
Capital Costs (\$USD/year)	1588219
CO ₂ Captured (tonne/year)	1222079
Levelized Cost of Capture (\$USD/tonne)	56.93

Discussion

The results of this simulation demonstrate that a 90% post-combustion CO₂ capture rate is achievable at a reboiler duty of 4.147 GJ/tonne CO₂ using a 30 wt% MEA absorber-stripper loop modeled in ASPEN Plus. This reboiler duty is approximately 12% above the Esbjerg pilot benchmark of 3.7 GJ/tonne CO₂, a deviation consistent with the goals and level of detail achieved in this simulation in addition to the slightly elevated lean loading shown in the results.

The sensitivity analyses exemplify two different key engineering tradeoffs. First, Figure 3 confirms the fundamental relationship between regeneration energy and solvent quality — lower lean loading requires higher reboiler duty but improves absorption and allows a lower L/G. The lean loading result of 0.302 is slightly above the optimal range reported in literature, suggesting modest room for improvement through increased reboiler duty or optimized column internals. Secondly, Figure 2 demonstrates that a minimum L/G of approximately 2.7 is required to achieve 90% capture at the base case lean loading. This is consistent with values reported by Freguia and Rochelle for similar operating conditions (Freguia & Rochelle, 2003). The optimization block reduced reboiler duty to 178.1 MW while maintaining 91.4% capture, a modest improvement over the base case that confirms the simulation is operating near its energy minimum within the confines of the simulation specifications.

The screening-level TEA yields a levelized cost of capture of \$56.93/tonne CO₂, which compares well to the NETL benchmark of \$58.2/tonne CO₂ for a supercritical pulverized coal plant with 90% CO₂ capture (Chou et al., 2015). Steam cost dominates operating expenditure, consistent with the findings of literature that reboiler energy penalty represents the primary economic driver for amine capture processes. The reboiler costing error shown in ASPENS economic analysis tool suggests the reported capital cost may be slightly underestimated, and actual plant costs would likely be modestly higher and more consistent with the NETL benchmark.

Conclusion

This work presents a steady-state rate-based simulation of a 90% post-combustion CO₂ capture system using 30 wt% MEA solvent, developed in ASPEN Plus and validated against the CASTOR Esbjerg pilot plant dataset. The simulation achieved 90.14% CO₂ capture at a reboiler duty of 4.147 GJ/tonne CO₂, within 12% of the Esbjerg benchmark. Sensitivity analysis quantified the tradeoff between reboiler duty and lean loading in addition to identifying the minimum L/G required for 90% capture. An optimization block confirmed that reboiler duty could be minimized to 178.1 MW while maintaining capture above 90%. Screening-level techno-economic analysis yielded a levelized cost of capture of \$56.93/tonne CO₂, consistent with NETL benchmark estimates for first-generation MEA capture

technology. The simulation, lab notes, raw data and associated Python analysis are publicly available at github.com/deankfullerton/co2-capture-simulation.

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